



Meal Logistics Planner: Automated Multi-Dish Cooking Scheduling with AI Planning

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The Problem

Cooking **multiple dishes** at the **same time** is difficult because each recipe has step dependencies and shared kitchen resources (burners, ovens, prep space). Planning manually often causes conflicts or dishes finishing at different times.

Our system automatically generates a **cooking schedule** so that all dishes finish at the same serving time while respecting recipe order and kitchen resource limits.

Why It Matters

Real-world problems involve coordinating tasks with limited resources. For example:



Goal:
All dishes ready at 6:00 PM

Challenges:
🧐 Different cooking times
🔍 Shared kitchen resources
📝 Recipe step dependencies

Representation

Cooking tasks are modeled using PDDL

| Predicates |

```
burner_available(b)
oven_available(o)
step_completed(s)
```

| Actions |

```
cook_pasta
sear_steak
roast_veggies
```

| Constraints |

```
burner_capacity
oven_capacity
step_dependencies
```

Example: 3-Dish Meal

Input: Seared Steak, Creamy Pasta, Roasted Veggies

Goal: All Ready in 75 min

Kitchen: 4 burners, 2 ovens, 1 knife

Example: Timeline

Time	Action	Resource
0-5	preheat_oven	oven
25-30	chop_veggies	knife
30-60	roast_veggies	oven
50-62	cook_pasta	burner
60-68	sear_steak	burner
62-70	cream_sauce	burner
70-75	combine	-

Example: Unsat

🧐 **Scenario:** 5 dishes need burners simultaneously, but kitchen has only 4

🔍 **Diagnostic:** Peak burner usage 5 exceeds capacity 3 at minutes 40-50

💡 **Suggestions:**

- Reduce dishes
- Increase serve time
- Alternative methods

⚡ **Peak Usage:**

- Burners: 5/4 ❌
- Oven: 1/2 ✅

Performance VS LLM

Scenario	LLM	Our System
Temp Conflict	❌	✅ Detects
State Tracking	❌	✅ Maintains state
Dependencies	Pasta before boiling	✅ Enforces order
Resources	Ignores burner limit	✅ Validates capacity

Why Symbolic Reasoning Wins:

Guarantees correctness,
maintains state, respects physics

Planning & Scheduling

1 Planning

Fast Downward generates a valid action sequence.

2 Scheduling

Aligns the plan with the target serving time.

3 Validation

Checks resource usage and detects conflicts.

Try Our Demo !



Multi-dish cooking is logistically hard

❌ **Timing Chaos**
Dishes finish at different times

❌ **Resource Conflicts**
Oven temperature or burner limits

❌ **State Tracking**
Forgetting which tools are clean or dirty

Current Solutions Fail

📱 **Recipe Apps**
Treat each dish in isolation

?? **LLMs**
Often hallucinate impossible plans